

DIGITAL IMAGE PROCESSING

LECTURE # 4

DIGITAL IMAGE FUNDAMENTALS-II

Contact Information

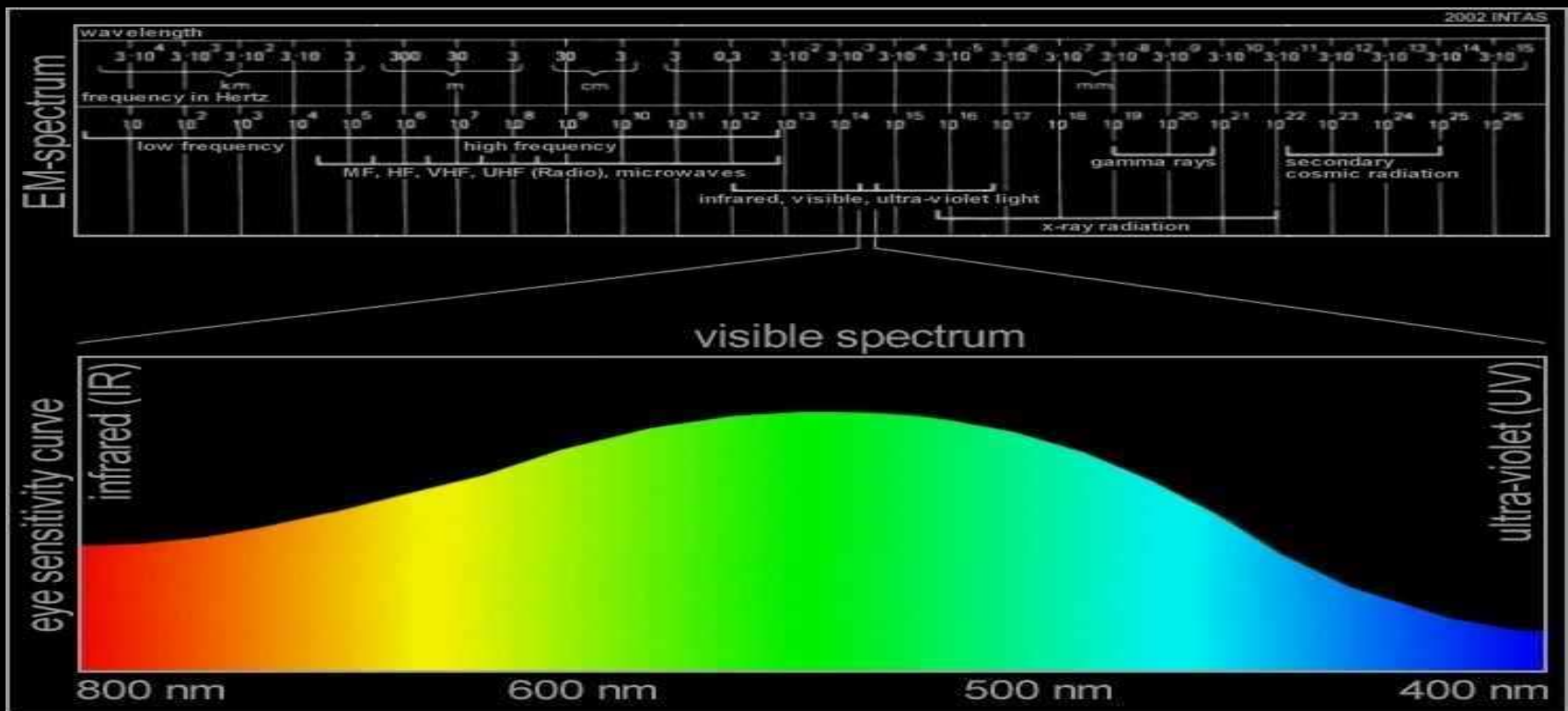
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Topics to Cover

- ❑ Light and EM Spectrum
- ❑ Visual Perception
- ❑ Structure Of Human Eyes
- ❑ Image Formation on the Eye
- ❑ Brightness Adaptation and Discrimination
- ❑ Optical Illusion
- ❑ Image Formation Model
- ❑ Image Acquisition using various sensor arrangements

Light and EM Spectrum



Light & EM Spectrum



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- ❑ Light is a particular type of EM radiation that can be seen by human eye.
- ❑ **Achromatic Light**
 - ✓ Light that is void of color is called achromatic or monochromatic light
 - ✓ The only attribute of such light is its intensity.
 - ✓ The term gray level generally is used to describe monochromatic intensity because it ranges from black to grays and finally to white
- ❑ **Chromatic light**
 - ✓ spans EM spectrum from 0.43 μm (violet) to 0.79 μm (red).
 - ✓ Three basic quantities are used to describe the quality of a chromatic light source
 1. **Radiance**
 2. **Luminance**
 3. **Brightness**

Light & EM Spectrum

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❑ Radiance

- ✓ The total amount of energy that flows from the light source
- ✓ Measured in Watts(W)

❑ Luminance

- ✓ Luminance, measured in lumens (lm), is a *measurable physical quantity* representing the amount of light emitted or reflected from a surface in a given direction.
- ✓ Luminance is measured in different units such as candela (cd/m²), footlambert (ftL), mililambert (mL), and Nit (nt).

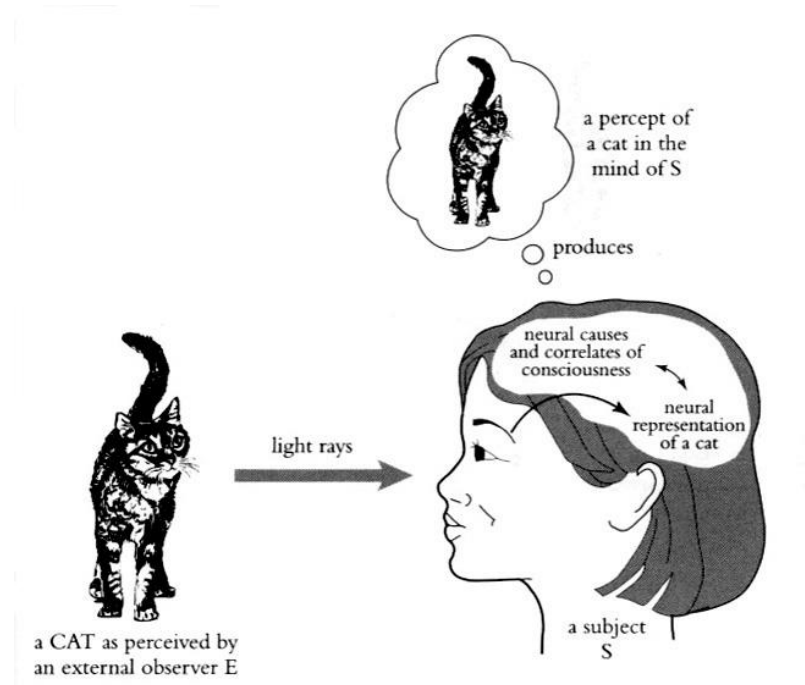
❑ Brightness

- ✓ Brightness refers to our subjective perception of how bright an object is. Therefore, what may seem very bright to you may be less bright to me. We can use subjective words such as *dim* and *very bright* to describe our perceptions of brightness.
- ❑ Luminance is the actual measurable light intensity. Brightness is how that luminance is perceived by the human eye. For example, a bright spotlight (high luminance) might seem less bright if you're in a brightly lit room, but in a dark room, the same spotlight would appear much brighter.

Visual Perception

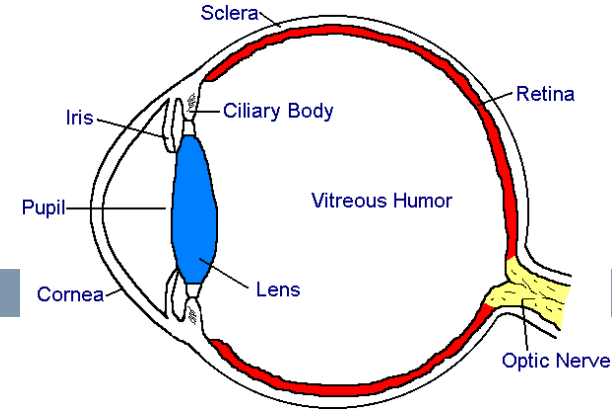
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- ❑ How are images formed in the eye?
- ❑ Eye's physical limitation?
- ❑ Human visual interpretation of images?



Structure of Human Eyes

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❑ Three membranes enclose the eye:

❑ Cornea and sclera

- ✓ Cornea is a tough, transparent tissue cover the anterior surface of the eye.
- ✓ Sclera is an opaque membrane enclose the remainder of the optic globe.

❑ Choroid

- ✓ A network of blood vessels for eye nutrition
- ✓ Heavily pigmented to reduce the amount of extraneous light entering the eye
- ✓ At its anterior extreme, it is divided into the *ciliary body* and *iris diaphragm*.
- ✓ Iris expands or contracts to control the amount of light entering the eye
- ✓ The central opening (the *pupil*) varies in diameter from 2 to 8 mm.

❑ Retina

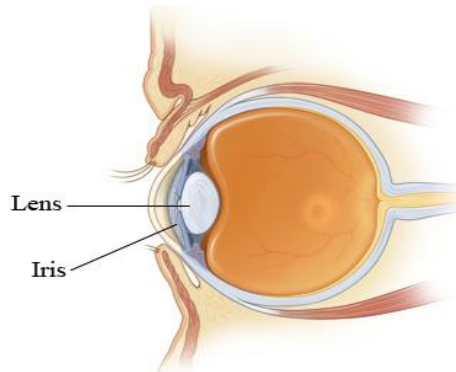
- ✓ Retina lines the insides of the wall's interior portion and connected by the optic nerve to the brain
- ✓ Distribution of two classes of receptors

Structure of Human Eyes

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□ Lens

- ✓ Lens is made of concentric layer of fibrous cells and is suspended by fiber that attached to the ciliary body.
- ✓ Colored by slightly yellow pigmentation that increases with age
- ✓ Excessive clouding of lens, called cataract, leads to poor color discrimination and loss of clear vision
- ✓ The lens absorbs approximately 8% of the visible light spectrum.
- ✓ The lens contains 60-70% water and 6% fat and protein.



Structure of Human Eyes – Eye Sensors

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- We see the scene with the sensors in the retina of the eye called rods and cones

Color Sensor

- Cones: (Red 65%, Green 33%, Blue 2%)
 - ✓ 6 – 7 millions located primarily in the central portion of the retina
 - ✓ Highly sensitive to color
 - ✓ Each one is connected to its own nerve end, allowing for fine resolution detail
 - ✓ Photopic or bright-light vision

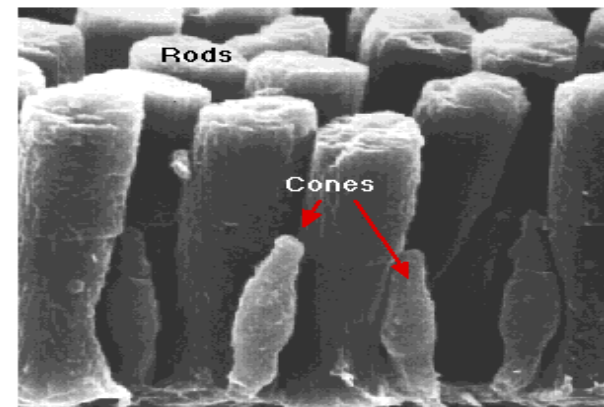
Brightness Sensor

- Rods
 - ✓ 75-150 millions distributed over the retinal surface with several rods connected to a single nerve end, providing coarse resolution
 - ✓ Not involved in color vision and sensitive to low illumination
 - ✓ Scotopic or dim vision

Structure of Human Eyes – Eye Sensors[Rods]

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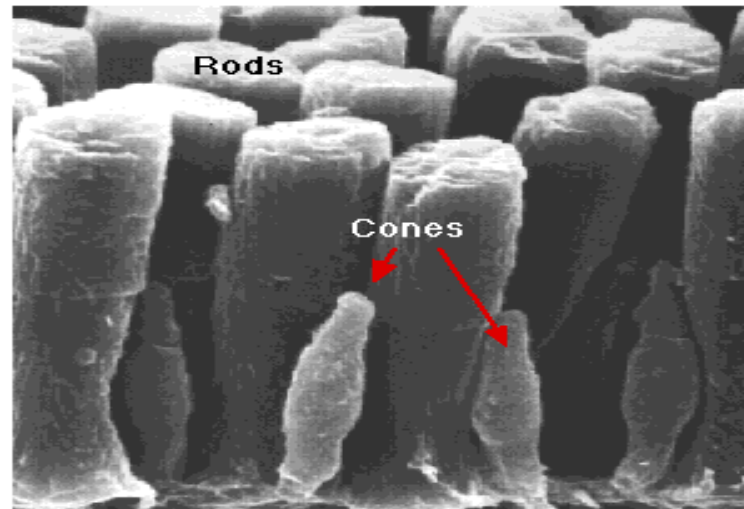
- ❑ Rods are more sensitive than the cones, but they are not sensitive to color, they perceive images as black, white and different shades of grey.
- ❑ They work well in dim light as they contain a pigment, rhodopsin, which is sensitive at low light intensity, but saturates at higher (Photopic) intensities.
- ❑ More than one thousand times as sensitive, the rods respond better to blue but very little to red wavelength
- ❑ Rods are distributed throughout the retina but there are none at the fovea and none at the blind spot. Rod density is greater in the peripheral retina than in the central retina.



Structure of Human Eyes – Eye Sensors[Cones]

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- ❑ Each cone contains one of three pigments sensitive to either RED GREEN or BLUE
- ❑ Each pigment absorbs a particular wavelength of color. There are short wavelength cones that absorb blue light, middle wavelength cones that absorb green light, and long wavelength cones that absorb red light
- ❑ When we observe a color that has a wavelength between that of the primary colors red, green and blue, combinations of the cones are stimulated.
- ❑ An example could be that yellow light stimulates cones that are sensitive to red and green light. The result is that we can detect light of all colors in the visible spectrum
- ❑ People who suffer color blindness have less numbers of cones than normal, so they get colors confused.



Structure of Human Eyes

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- ❑ The cones are most dense in the center of retina (fovea).
- ❑ Density of cones in the area of fovea is 150,000 element/mm²

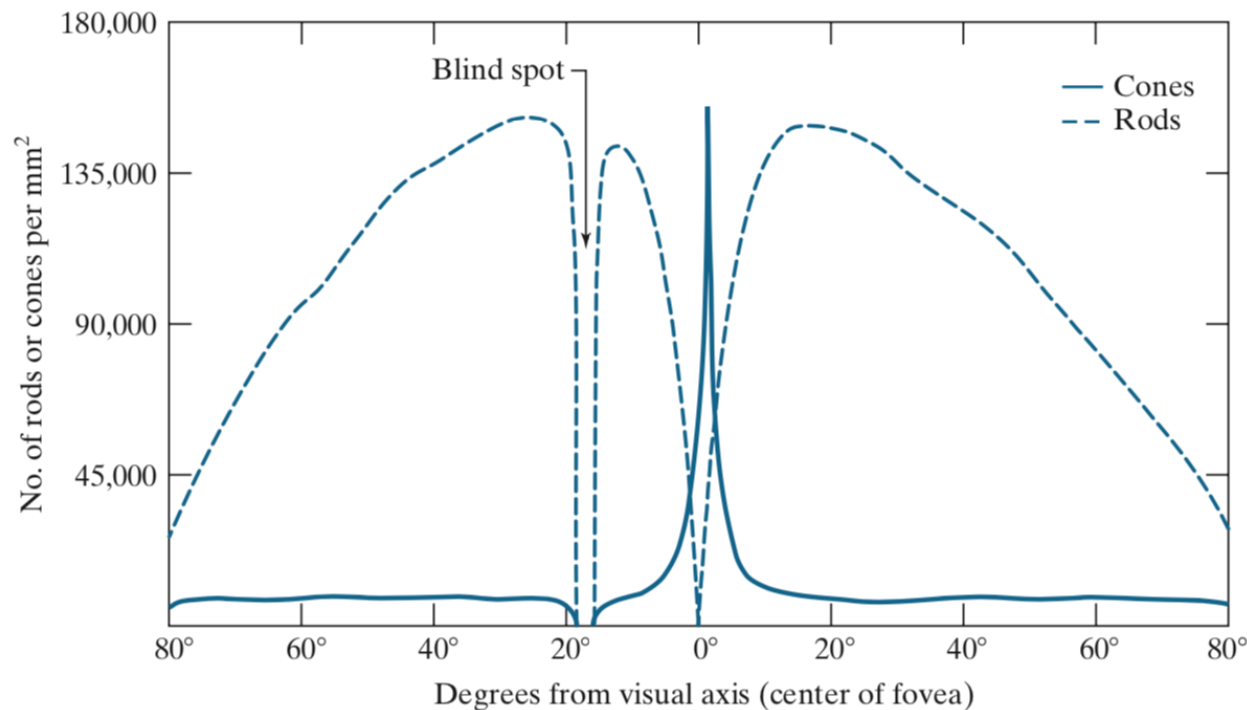


FIGURE 2.2
Distribution of rods and cones in the retina.

Structure of Human Eyes – Field of View [2]

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- ❑ The **field of view** (also **field of vision**, abbreviated **FOV**) is the extent of the observable world that is seen at any given moment.
- ❑ Different animals have different fields of view, depending on the placement of the eyes.
- ❑ Humans have an almost 180-degree forward-facing horizontal field of view, while some birds have a complete or nearly-complete 360-degree field of view. In addition, the vertical range of the field of view in humans is typically around 100 degrees.



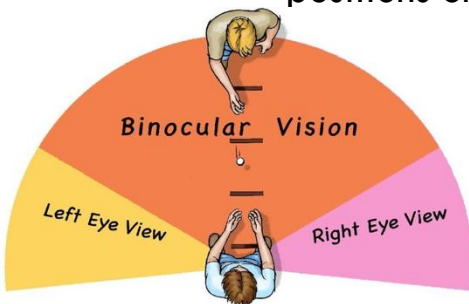
Field of View- Binocular vision[3]

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- ❑ Binocular vision is vision in which both eyes are used together. The word binocular comes from two Latin roots, bini for double, and oculus for eye
- ❑ Having two eyes confers at least four advantages
 - ✓ First, it gives a creature a spare eye in case one is damaged.
 - ✓ Second, it gives a wider field of view. For example, humans have a maximum horizontal field of view of approximately 200 degrees with two eyes, approximately 120 degrees of which makes up the binocular field of view (seen by both eyes) flanked by two unocular fields (seen by only one eye) of approximately 40 degrees
 - ✓ Third, it gives binocular summation in which the ability to detect faint objects is enhanced.
 - ✓ Fourth it can give stereopsis [4] in which parallax provided by the two eyes' different positions on the head give precise depth perception [5]

Depth perception is the visual ability to perceive the world in three dimensions (3D) and the distance of an object.

Stereopsis (from stereo- meaning "solid" or "three-dimensional", and opsis meaning appearance or sight) is the impression of depth that is perceived when a scene is viewed with both eyes by someone with normal binocular vision.



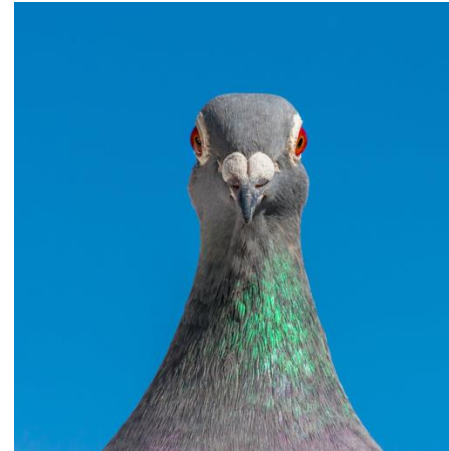
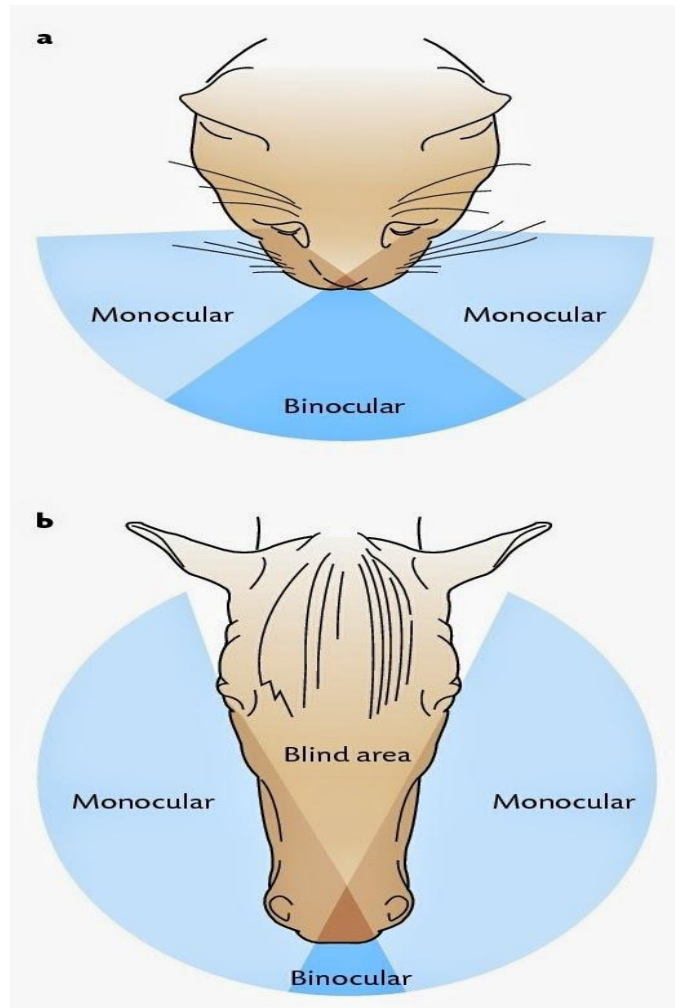
Field of View- Monocular vision^[4]

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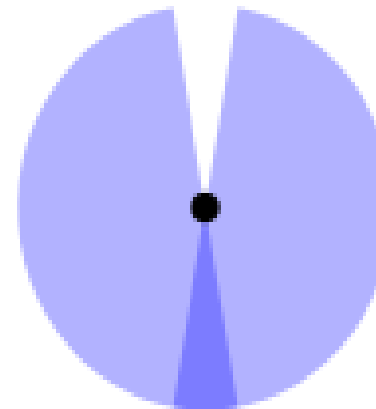
- ❑ **Monocular vision** is vision in which each eye is used separately. By using the eyes in this way, as opposed by binocular vision, the field of view is increased, while depth perception is limited.
- ❑ The eyes are usually positioned on opposite sides of the animal's head giving it the ability to see two objects at once. The word monocular comes from the Greek root, *mono* for one, and the Latin root, *oculus* for eye.
- ❑ Most birds and lizards (except chameleons) have monocular vision.
- ❑ Owls and other birds of prey are notable exceptions. Also, many prey have monocular vision to see predators.

Field of View- Monocular vision & Binocular Vision

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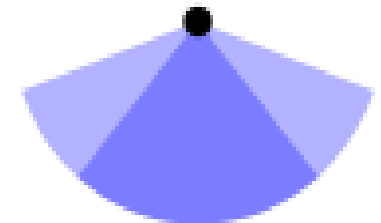
Pigeon



■ Binocular vision



Owl



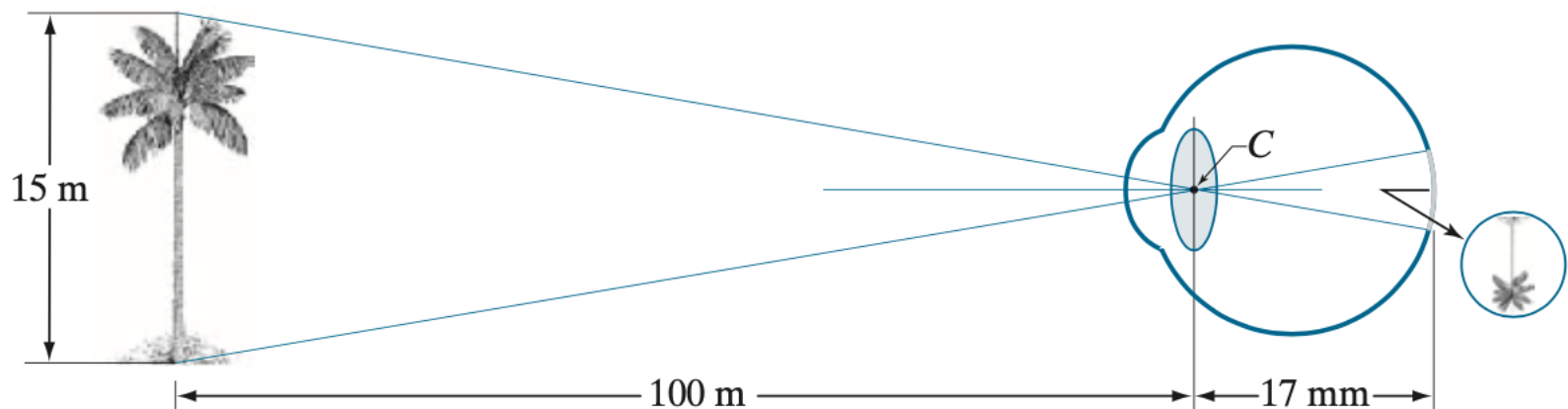
■ Monocular vision



Image Formation in the Eyes

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- The distance between the center of the lens and the retina (focal length) varies from 17mm to 14mm.
- The tension of fibers of the ciliary body controls the lens shape.
- The retinal image is reflected primarily in the fovea.
- Perception = excitation of light receptors, which transform radiant energy into electrical impulses ultimately decoded by the brain.
- If the tree is 100m away from a distance having an actual height of 50m and h is the height in mm of the tree in the retinal image, then calculate h ?



Brightness Adaptation & Discrimination

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- ❑ The range of light intensity levels to which the human visual system can adapt is enormous – on the order of 10^{10} .
- ❑ The *subjective brightness* is a logarithmic function of light intensity incident on the eye.
- ❑ In photopic vision, the range is about 10^6 .
- ❑ The current sensitivity level it can discriminate simultaneously is rather small compared with the total adaptation range
- ❑ *Brightness adaptation level*: the current sensitive level of the visual system.

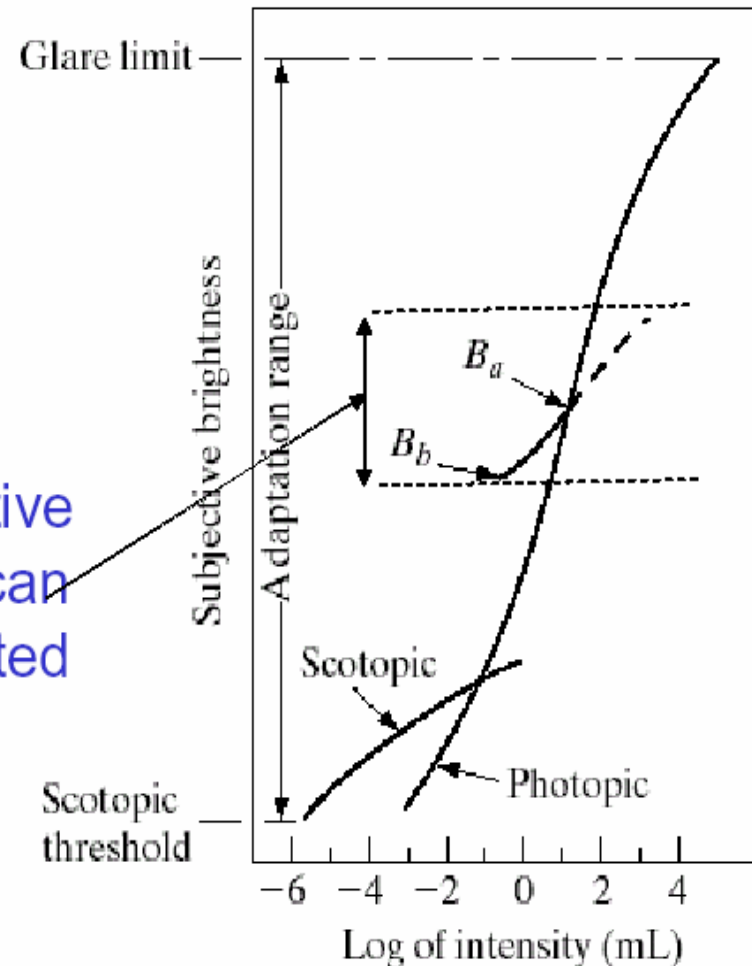
Brightness Adaptation & Discrimination

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FIGURE 2.4

Range of subjective brightness sensations showing a particular adaptation level.

The range of subjective brightness that eye can perceive when adapted to this level



Brightness Adaptation & Discrimination

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Experiments: Apply a short-duration flash at a circle to see if ΔI is bright enough

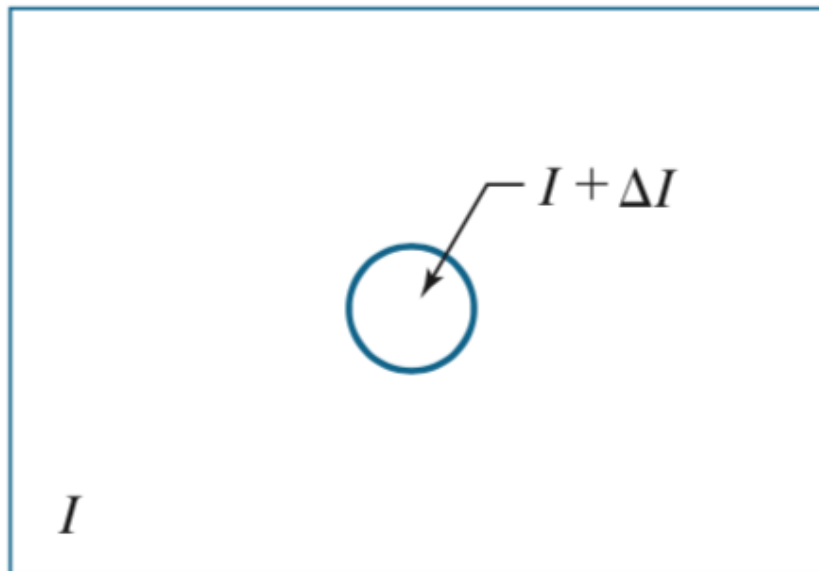


FIGURE 2.5 Basic experimental setup used to characterize brightness discrimination.

Brightness Adaptation & Discrimination

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- ❑ The dlc is the increment of illumination discriminable 50% of the time with the background illumination I .
- ❑ The quantity dlc/I is called the Weber ratio.
- ❑ The *smaller* dlc/I means that a small percentage change in intensity is discriminable – *good brightness discrimination*
- ❑ If the background illumination is constant, the intensity of object is allowed to vary incrementally from never perceived to always being perceived.
- ❑ Typically, the observer can discern a totally from one to two dozens different intensity changes.



Brightness Adaptation & Discrimination

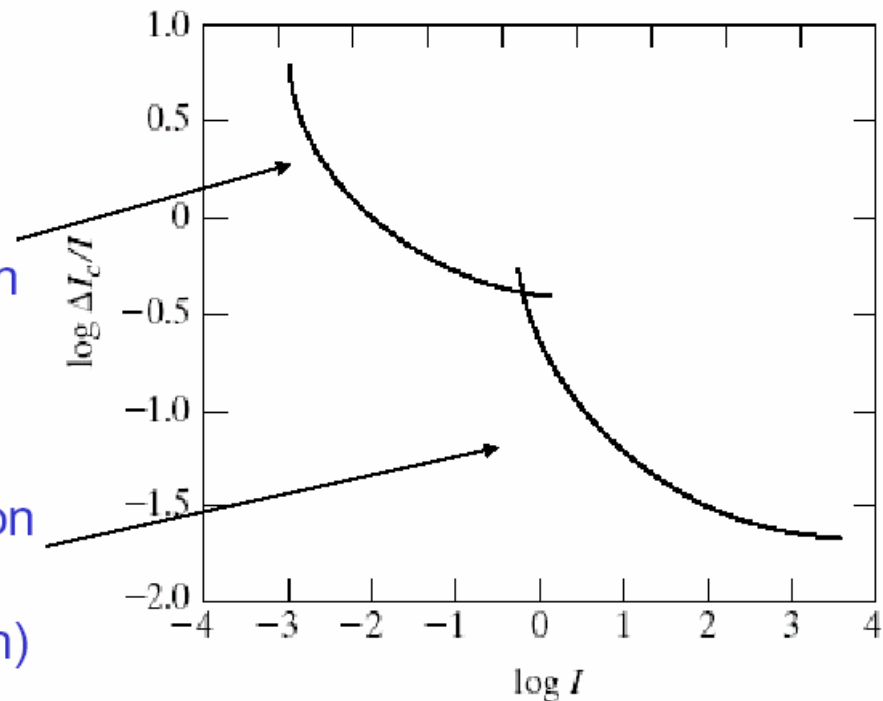
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- Perceived brightness is not a simple function of intensity, rather it is log of intensity and it is described by two phenomena's
 - ✓ Mach Band Effect
 - ✓ Simultaneous Contrast

FIGURE 2.6
Typical Weber
ratio as a function
of intensity.

Low-level illumination
vision (rod cells)

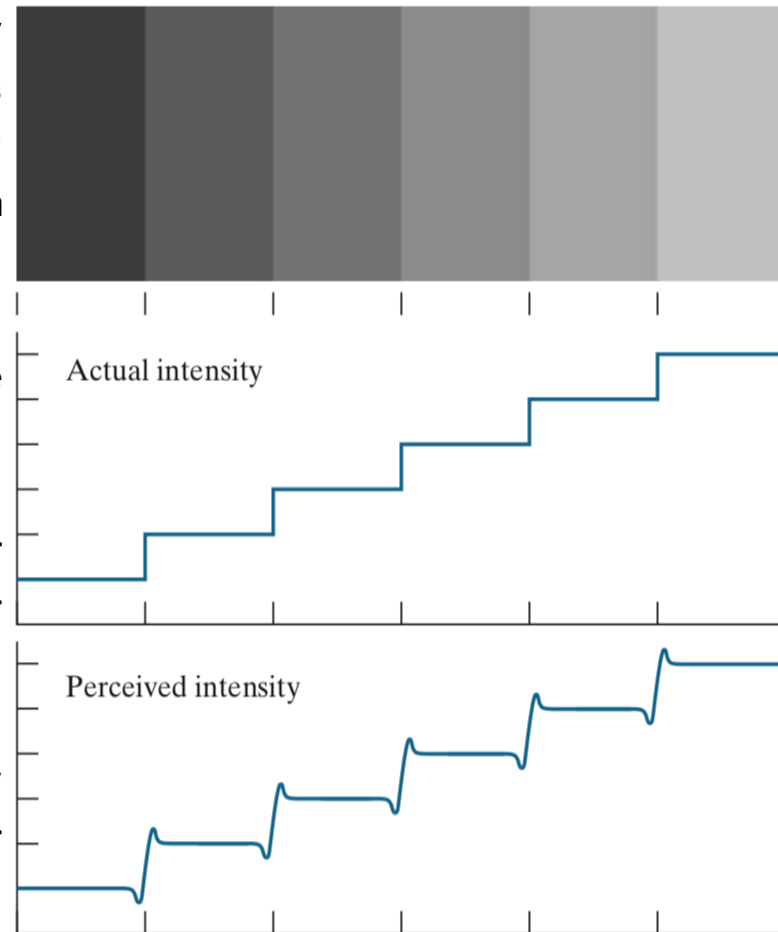
High-level illumination
vision (cone cells)
(better discrimination)



Mach Band Effect

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- ❑ **Mach bands** is an **optical illusion** named after the physicist **Ernst Mach**. It exaggerates the **contrast** between edges of the slightly differing shades of gray, as soon as they get in contact with each other
- ❑ Scalloped brightness pattern near the boundaries shown in stripes of constant intensity
- ❑ Visual system tends to under or overshoot around the boundary of two regions of different intensity.
- ❑ In this image, edges between bars appear brighter on the right side and darker on the left side.



Brightness Adaptation & Discrimination

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- A region's perceived brightness does not simply depend on its intensity but also on the background – *Simultaneous contrast*.



a b c

FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

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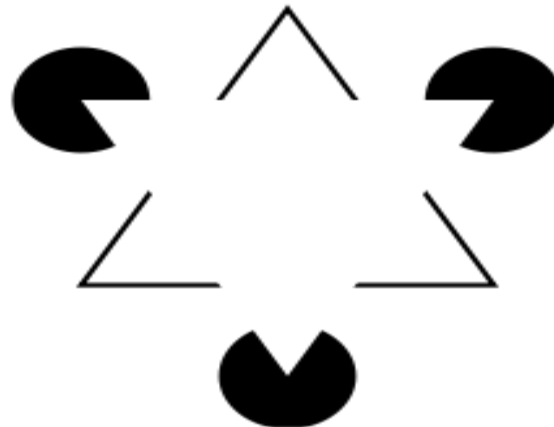
OPTICAL ILLUSION



Optical illusion [7]

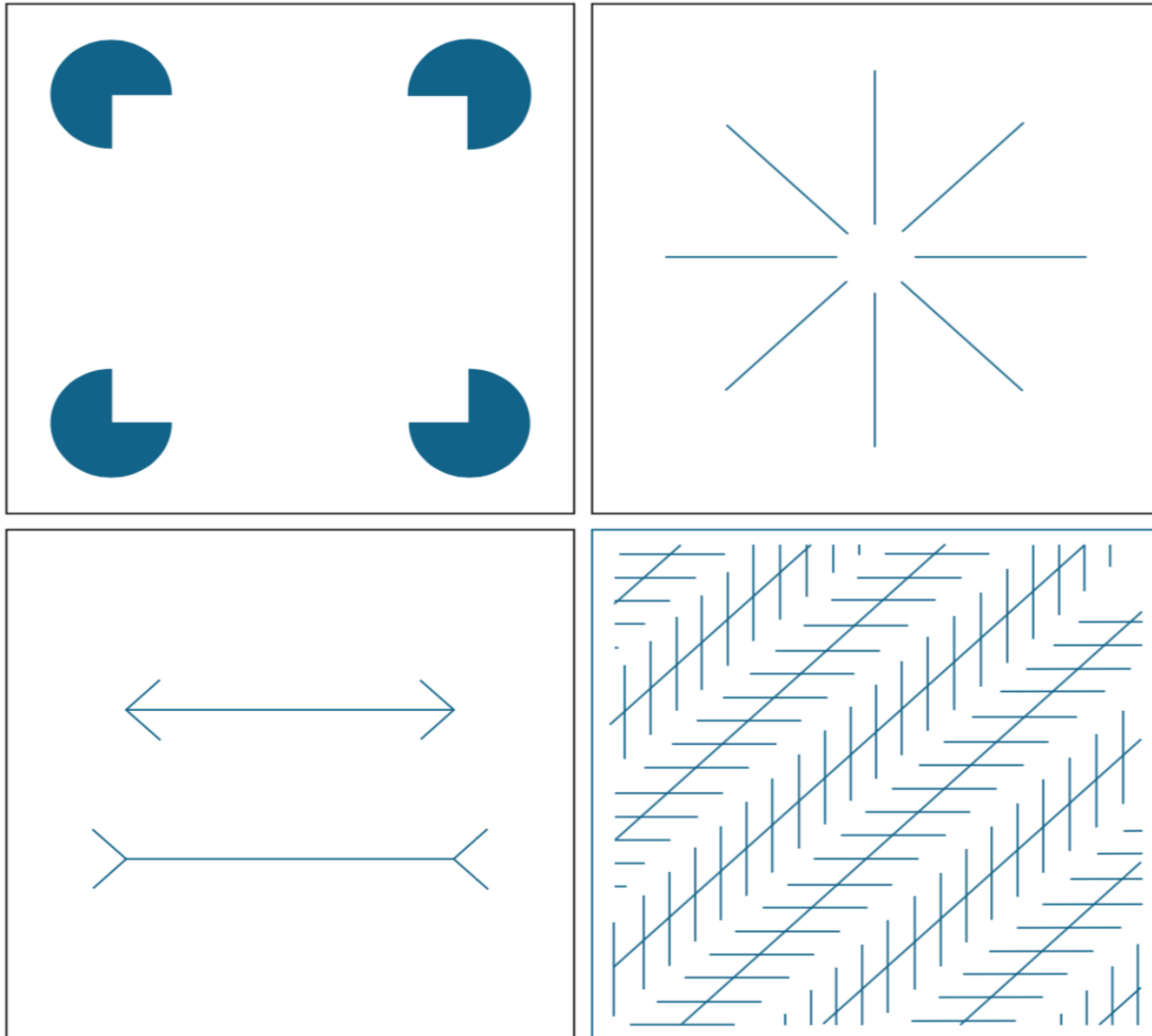
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- ❑ An **optical illusion** (also called a **visual illusion**) is characterized by visually perceived images that differ from objective reality.
- ❑ The information gathered by the eye is processed in the brain to give a perception that does not tally with a physical measurement of the stimulus source.
- ❑ Optical illusion is a phenomena in which the eye fills in non existing information or wrongly perceives the geometrical properties of objects



Optical illusion

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Optical illusion

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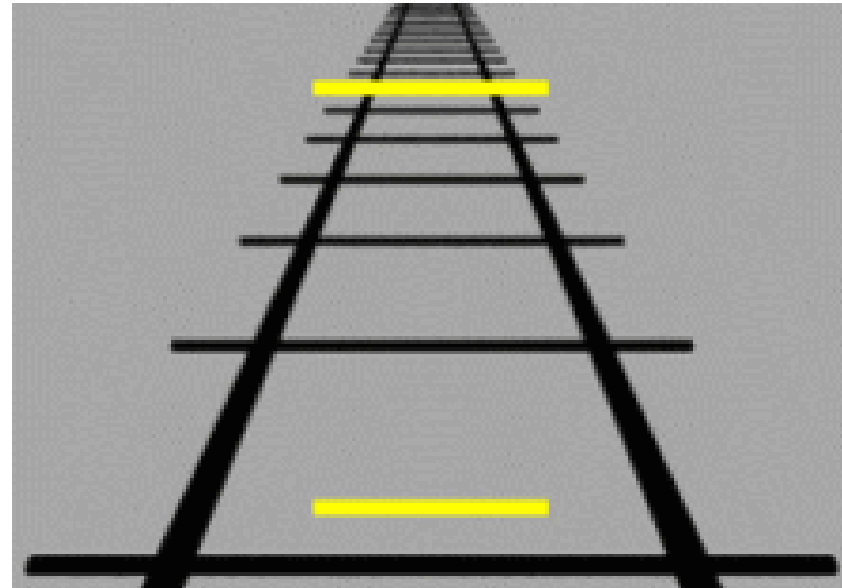


Old Woman...Or Young Girl?



Optical illusion

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Optical illusion

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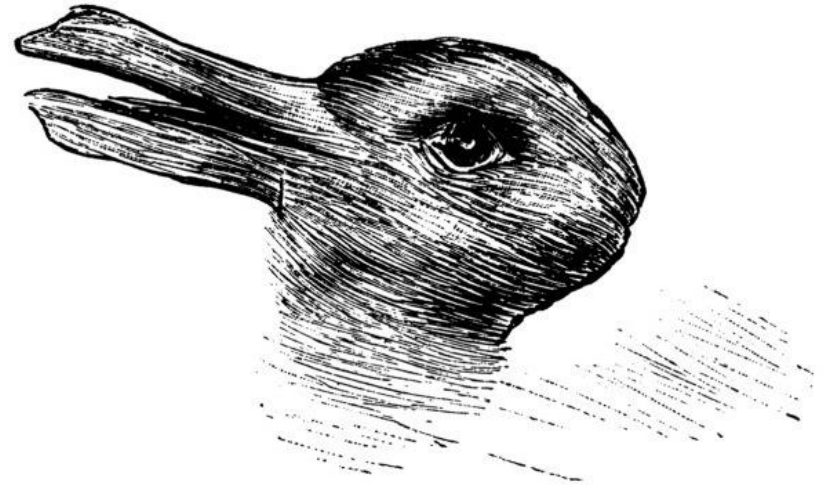
Optical illusion

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IF YOU SEE
2 IMAGES
IN 15s,
YOU ARE
A GENIUS!



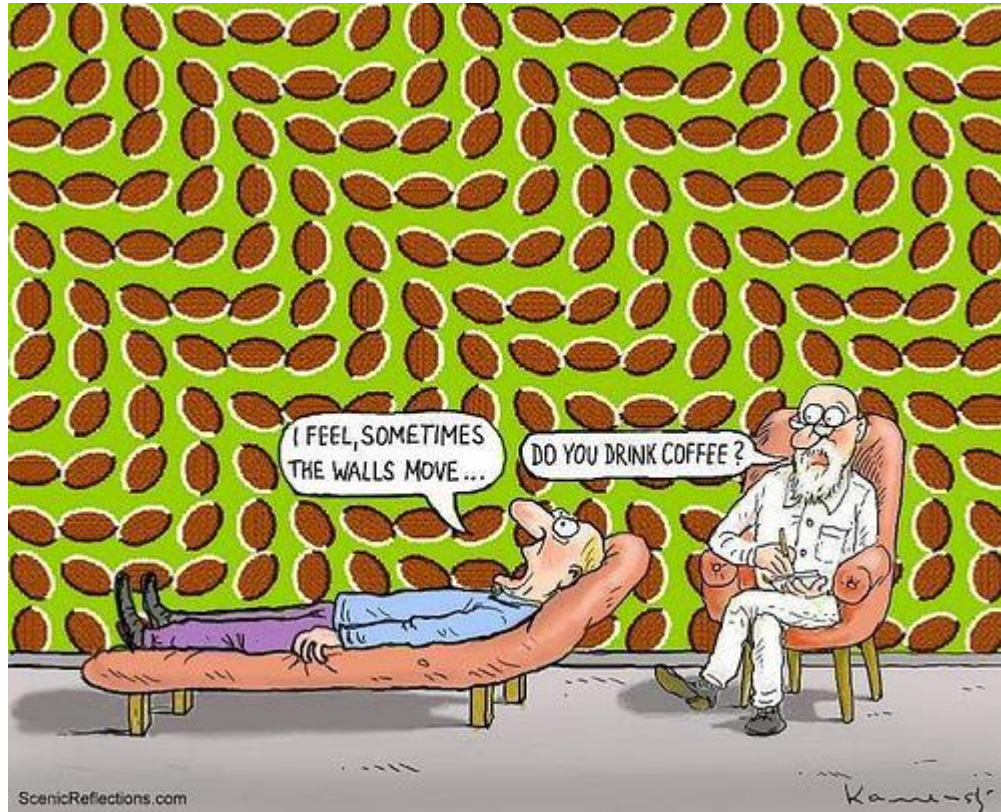
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Do you see a duck or a rabbit?

Optical illusion

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Optical illusion

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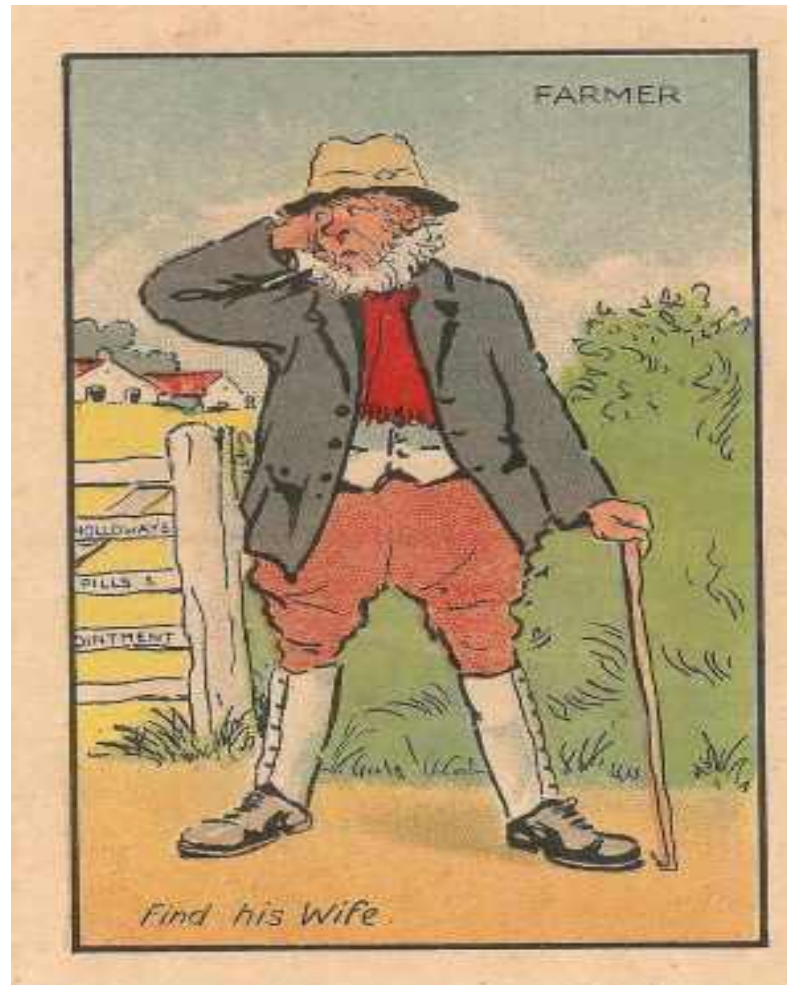


Image Acquisition

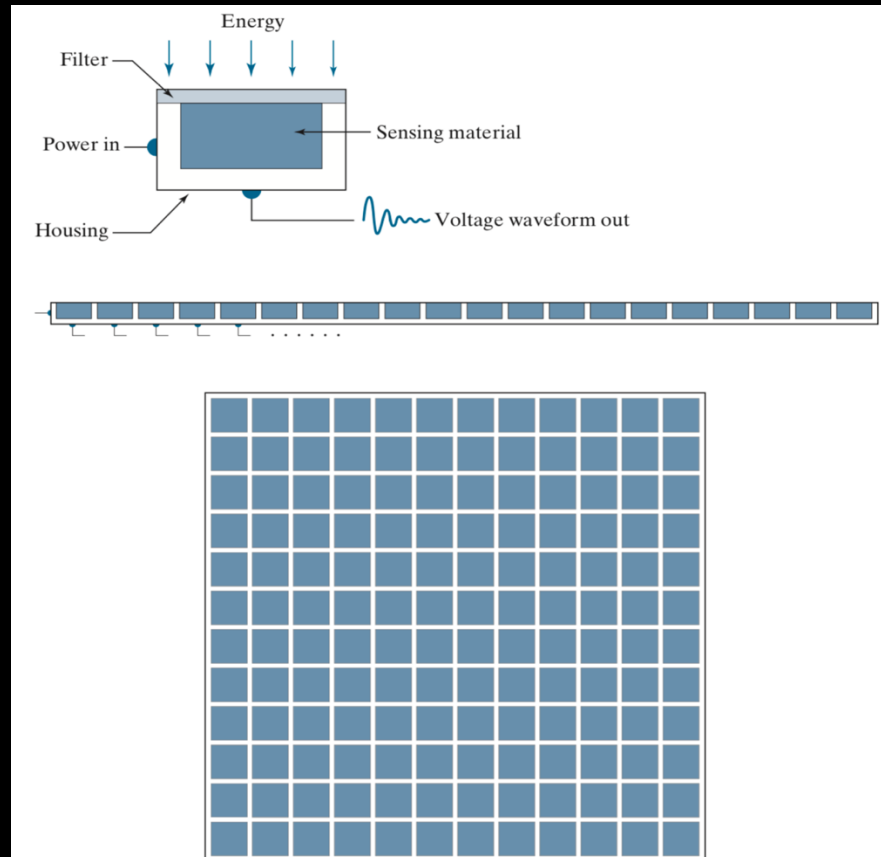
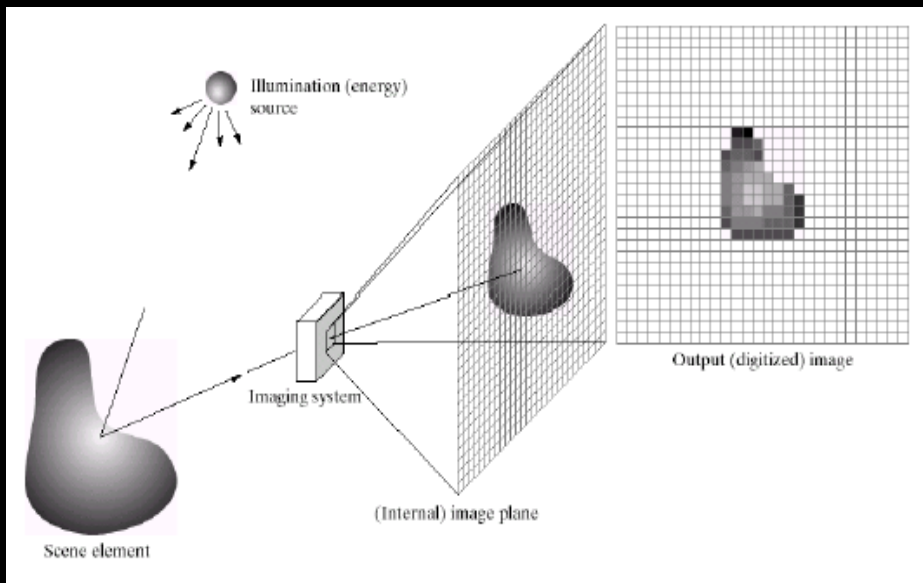


Image Acquisition using Point Sensor

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- ❑ Specify the location of vertical and horizontal motors
- ❑ Sense the light reflection
- ❑ Voltage waveform will be received (Analog signal)
- ❑ Convert this analog signal into digital signal through sampling and quantization
 - ✓ Apply Sampling to digitize coordinate values
 - ✓ Apply Quantization to digitize amplitude values
- ❑ Store the digitized value in memory

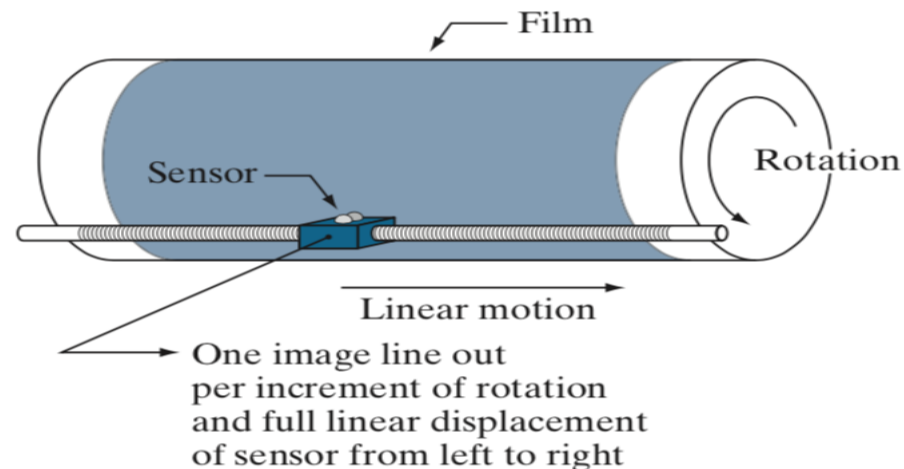
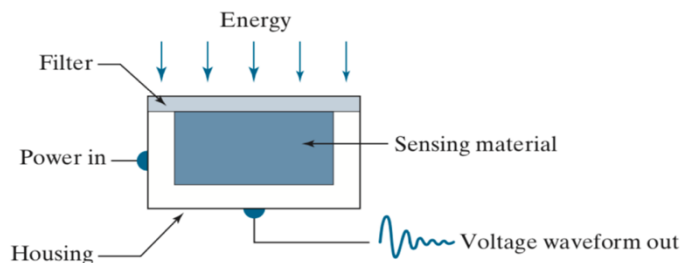


Image Acquisition using Line Sensor

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- ❑ Specify the location of vertical motor
- ❑ Sense the light reflection
- ❑ Voltage waveform will be received (Analog signal)
- ❑ Convert this analog signal into digital signal through sampling and quantization
 - ✓ Apply Sampling to digitize coordinate values
 - ✓ Apply Quantization to digitize amplitude values
- ❑ Store the digitized value in memory

Line sensor

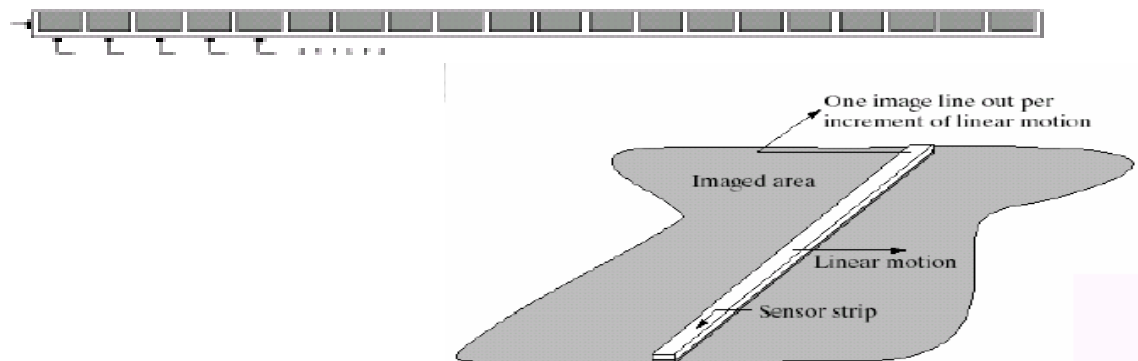


Image Acquisition using Array Sensor

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- ❑ Figure shows individual sensors arranged in a form of 2-D array
- ❑ This arrangement exists in modern day digital cameras
- ❑ A typical sensors for these cameras is a CCD array, which can be manufactured with a broad range of sensing properties and can be packaged in arrays of 4000 x 4000 elements or more
- ❑ CCD sensors are used widely in digital cameras and other light sensing instruments
- ❑ The response of each sensor is proportional to the integral of the light energy projected on to the surface of the sensor

Array sensor

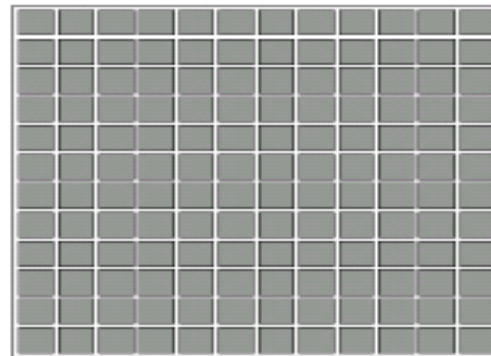
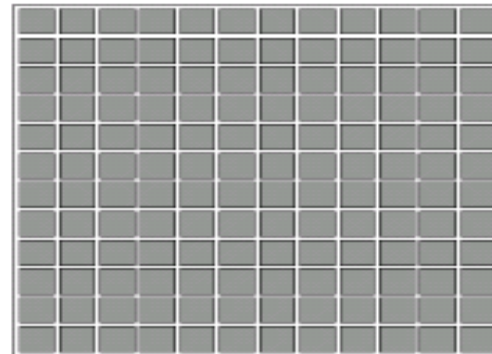


Image Acquisition using Array Sensor

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- ❑ Sense the light reflection on the sensor (arranged in 2D form)
- ❑ Voltage waveform will be received (Analog signal)
- ❑ Convert this analog signal into digital signal through sampling and quantization
 - ✓ Apply Sampling to digitize coordinate values
 - ✓ Apply Quantization to digitize amplitude values
- ❑ Store the digitized value in memory

Array sensor



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Image Formation Model

Image Formation Model

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Object Visibility

- ❑ Object becomes visible when illuminating source strikes the objects and due to reflection, our eyes can see the object because reflection reaches our eye after striking through object
- ❑ Scene Visibility = Reflection from the object , Light Source
- ❑ Image = reflectance , illumination



Image Formation Model

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- For monochromatic image 2-D array: $f(x, y)$
- The $f(x, y)$ is characterized by two components:
 - The amount of **source illumination** incident on the scene, i.e., $i(x, y)$.
 - The amount of illumination reflected by the objects in the scene, i.e., **reflectivity** $r(x, y)$.

- $f(x, y) = i(x, y) r(x, y)$
where $0 < i(x, y) < \infty$ and $0 < r(x, y) < 1$

- Reflectivity function: $r(x, y)$
- For X-ray, **transmissivity** function
- The **intensity** of monochrome image is

$$L_{\min} \leq f(x, y) \leq L_{\max} \quad L_{\min} = i_{\min} r_{\min} \quad \text{and} \quad L_{\max} = i_{\max} r_{\max}$$

- The interval $[L_{\min}, L_{\max}]$ is called the gray scale
- Common practice is to shift this interval to $[0 \text{ to } L-1]$, where $L=0$ is considered black and $L=L-1$ is considered white on the gray scale
- All intermediates are shades of gray varying from black to white

References

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1. DIP by Gonzalez
2. http://en.wikipedia.org/wiki/Field_of_view
3. http://en.wikipedia.org/wiki/Binocular_vision
4. http://en.wikipedia.org/wiki/Monocular_vision
5. <http://en.wikipedia.org/wiki/Stereopsis>
6. http://en.wikipedia.org/wiki/Depth_perception
7. http://en.wikipedia.org/wiki/Optical_illusion
8. <http://www.michaelbach.de/ot/>

For any query Feel Free to ask



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